COHOMOLOGY OF PARASITIC MODULES VANISHES AND PUSHFORWARDS OF PARASITIC MODULES ARE PARASITIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Pushforward and restriction formula. Sheaves on Stacks, Lemma stacks-sheaves-lemma-pushforward-restriction. Describes R^i f_* F as a sheafification of stalkwise cohomology.
- Smooth covering computes cohomology. Sheaves on Stacks, Proposition stacks-sheaves-proposition-smooth-covering-compute-cohomology. Spectral sequence used twice to reduce the stack to a scheme.
- Cohomology of parasitic modules on schemes. Descent, Lemma descent-lemma-cohomology-parasitic. Covers the final scheme-representable case.

Source reference. Original reference 075G: sources/stacks-chow/tags.txt, source lines 9130–9130. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0774: sources/stacks-chow/tags.txt, source lines 9189–9189. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06XJ: sources/stacks-chow/tags.txt, source lines 8853–8853. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 076Z: sources/stacks-chow/tags.txt, source lines 9184–9184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0755: sources/stacks-chow/tags.txt, source lines 9119–9119. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\tau \in \{étale, fppf\}$. Let $\mathcal{X}$ be an algebraic stack. Let $\mathcal{F}$ be a parasitic object of Mod$(\mathcal{X}_\tau, \mathcal{O}_\mathcal{X})$.*

(1) *$H^i_\tau(\mathcal{X}, \mathcal{F}) = 0$ for all i.*
(2) *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Then $R^i f_* \mathcal{F}$ (computed in τ-topology) is a parasitic object of Mod$(\mathcal{Y}_\tau, \mathcal{O}_\mathcal{Y})$.*

Proof. We first reduce (2) to (1). By Sheaves on Stacks, Lemma 075G we see that $R^i f_* \mathcal{F}$ is the sheaf associated to the presheaf

$$y \longmapsto H^i_\tau\Big(V \times_{y,\mathcal{Y}} \mathcal{X},\ \mathrm{pr}^{-1}\mathcal{F}\Big)$$

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Here y is a typical object of $\mathcal{Y}$ lying over the scheme V. By Lemma 0774 it suffices to show that these cohomology groups are zero when $y : V \to \mathcal{Y}$ is flat. Note that $\mathrm{pr} : V \times_{y,\mathcal{Y}} \mathcal{X} \to \mathcal{X}$ is flat as a base change of y. Hence by Lemma 0774 we see that $\mathrm{pr}^{-1}\mathcal{F}$ is parasitic. Thus it suffices to prove (1).

To see (1) we can use the spectral sequence of Sheaves on Stacks, Proposition 06XJ to reduce this to the case where $\mathcal{X}$ is an algebraic stack representable by an algebraic space. Note that in the spectral sequence each $f_p^{-1}\mathcal{F} = f_p^*\mathcal{F}$ is a parasitic module by Lemma 0774 because the morphisms $f_p : \mathcal{U}_p = \mathcal{U} \times_{\mathcal{X}} \ldots \times_{\mathcal{X}} \mathcal{U} \to \mathcal{X}$ are flat. Reusing this spectral sequence one more time (as in the proof of Lemma 076Z) we reduce to the case where the algebraic stack $\mathcal{X}$ is representable by a scheme X. Then $H^i_\tau(\mathcal{X}, \mathcal{F}) = H^i((Sch/X)_\tau, \mathcal{F})$. In this case the vanishing follows easily from an argument with Čech coverings, see Descent, Lemma 0755. $\square$